The Smallest Circle Touching Three Lines in 3D

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Abstract

We solve Tim Wylie's problem of the smallest circle in $\mathbb{R}^3$ that touches three lines. It is a great circle of a smallest ball that meets all three lines. An $O(1)$ test decides whether two or three lines are active. We give closed forms for 10 of the 16 cases of the problem's Figure 1. For the other 6, only the two-active subcase is closed. A numerical solver agrees wherever a closed form exists.

1 Introduction

Given three lines in $\mathbb{R}^3$, we want the circle of smallest radius that touches all of them (UTRGV Research Hackathon 2026, Tim Wylie). We return its center c, radius r and plane normal n, for all positions in the problem's Figure 1. The ball subproblem is the intersection radius problem [1]. We add the circle-to-ball reduction, an active-line test and closed forms. We read "touch" as "intersect": the circle curve shares a point with each line. We do not treat the tangent reading. A line is $L_i = \{a_i + t\, d_i\}$ with $|d_i| = 1$, and $P_i = I - d_i d_i^{\top}$. The distance from c to L_i is $\mathrm{dist}_i(c) = |P_i(c - a_i)|$, and the foot is $q_i = a_i + ((c - a_i) \cdot d_i)\, d_i$.

2 Main Theorem

Let $f(c) = \max_i \mathrm{dist}_i(c)$ and $r^* = \min_c f(c)$. The ball $B(c^*, r^*)$ is a smallest ball that meets all three lines.

Lemma 1 (Lower bound)**.** *A circle with center c and radius r that touches all three lines has $r \geq f(c) \geq r^*$.*

Proof. Let p_i be a common point of the circle and L_i. Then $\mathrm{dist}_i(c) \leq |c - p_i| = r$. □

Lemma 2 (Attainment)**.** *Some c^* attains r^*, and a great circle of $B(c^*, r^*)$ touches all three lines.*

Proof. Each dist_i is a norm of an affine map, so f is convex. If two lines are not parallel, $f \to \infty$ at infinity and a minimizer exists. If all are parallel, we reduce to the cross-section. If $r^* = 0$, the circle is the common point of the lines. Let $r^* > 0$ and $u_i = (c^* - q_i)/r^*$. Line i is *active* if $\mathrm{dist}_i(c^*) = r^*$. By the subdifferential of a maximum of convex functions, $0 \in \mathrm{conv}\{u_i : i \text{ active}\}$. The active u_i are unit vectors, so at least two lines are active.

Three active. The u_i are coplanar, so the feet $q_i = c^* - r^* u_i$ lie in a plane through c^*, which gives a great circle.

Two active, i and j. Then $u_i = -u_j$, so q_i, q_j are antipodal. Since $\mathrm{dist}_k(c^*) \leq r^*$ and L_k is unbounded, L_k meets the sphere at some s. The plane through q_i, q_j, s contains the diameter $q_i q_j$, hence c^*. □

Theorem 1. *The smallest touching circle has radius r^* and is a great circle of a smallest ball meeting all three lines.*

Proof. Lemma 1 gives $r \geq r^*$, and Lemma 2 attains it. A circle of radius r^* about c' needs $f(c') \leq r^*$, so c' minimizes f. □

Dispatcher. For each pair (i, j) let $\rho_{ij} = \mathrm{dist}(L_i, L_j)/2$. Let C_{ij} (a point, midline or line) be the centers of the smallest ball meeting L_i, L_j. Take the largest ρ_{ij}, with third line L_k. If the largest value is tied, test every tied pair. If $\mathrm{dist}(C_{ij}, L_k) \leq \rho_{ij}$, two lines are active, $r = \rho_{ij}$, and $c \in C_{ij}$ is nearest to L_k. The plane holds the diameter $q_i q_j$ and a point $s \in L_k$ on the sphere (Lemma 2). Otherwise three are active (Table 1). The test is closed form and $O(1)$.

Lemma 3 (Dispatcher)**.** *If every pair with the largest ρ fails the test, three lines are active. If a pair passes, $r^* = \rho_{ij}$.*

Proof. Let only i, j be active. Then $u_i = -u_j$, so $q_j - q_i = 2(c^* - q_i)$. This vector is normal to d_i and d_j, so $q_i q_j$ is the common perpendicular. Hence $2r^* = \mathrm{dist}(L_i, L_j)$, that is, $r^* = \rho_{ij}$. Also $r^* \geq \rho_{kl}$ for every pair, so ρ_{ij} is the largest. Further $c^* \in C_{ij}$, and $\mathrm{dist}(C_{ij}, L_k) \leq \mathrm{dist}_k(c^*) \leq r^* = \rho_{ij}$, so the pair passes. The contrapositive is the first claim. If a pair passes, a ball of radius ρ_{ij} meets all three lines, so $r^* \leq \rho_{ij} \leq r^*$. □

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Figure 1: (a) Smallest ball, great circle. (b) Obtuse 3-parallel case. Why: the plane may be tilted.

Case	Active	Answer: c, r, plane	Status
3 meet (triangle ABC)	3	Incircle, in the triangle's plane. $c = (\ell_a A + \ell_b B + \ell_c C)/(\ell_a + \ell_b + \ell_c)$, $r = 2\,\text{Area}/(\ell_a + \ell_b + \ell_c)$, with $\ell_a = \lvert BC\rvert$ etc.	closed
Concurrent or coincident (not in problem Fig. 1)	0 or 2	Concurrent: $r = 0$, $c = P$, a point circle. Coincident: two lines, r is half their distance.	closed
2 meet, 1 cross	2 or 3	If the line meeting both passes within $\delta/2$ of the midpoint m of the skew pair's common perpendicular: $c = m$, $r = \delta/2$. Otherwise three are active (open).	2-active closed
1 meet, 2 cross (over, over/under)	2 or 3	Dispatcher decides. Label: side of the meeting pair's plane on which L_3 passes.	2-active closed
0 meet (all skew)	2 or 3	Two active: dispatcher. Three active (open): quadrics $\lvert P_i(c - a_i)\rvert^2 = r^2$ and the cubic $\det[q_1 - c, q_2 - c, q_3 - c] = 0$. Bézout bound 24 in the unknowns (c, r). The system is even in r, so at most 12 solutions have $r > 0$.	2-active closed
2 parallel, 1 meets both or 1 meet 1 cross	2	$r = w/2$. For "meets both", c is any point of M within $w/2$ of the transversal.	closed
2 parallel, 0 meet (over, over/under)	2 or 3	Dispatcher first. If (L_1, L_2) has the largest ρ and $\text{dist}(M, L_3) \le h$, then $r = h$. If another pair has the largest ρ and passes, then r is its ρ. If three are active, $x = 0$ and $r^2 = h^2 + z^2$. Here z is the root of $q(z) = Az^2 + Bz + C = h^2$ of smallest $\lvert z\rvert$. With $k = e_x a_{3,z} - e_z a_{3,x}$ and $D = 1 - e_y^2$: $A = -e_z^2/D$, $B = -2e_x k/D$, $C = k^2/D$.	closed
3 parallel, coplanar	2	r is half the gap of the outer two lines. c is any point of their midline.	closed
3 parallel, 2 coplanar (prism)	3 or 2	Cross-section triangle T. Acute: circumcircle of T, plane $\perp$ lines. Right or obtuse: $r =$ longest side/2, plane tilted (our Fig. 1b).	closed
1 meet + perpendicular	2 or 3	3 active: $c = P + tb$, $r = t\sin(\theta/2)$, $t =$ smaller root of $t^2\cos^2(\theta/2) - 2t\,(b\cdot v) + \lvert v\rvert^2 = 0$.	closed
1 cross + perpendicular	2 or 3	A 180° rotation swapping L_1, L_2 may remain. If S is on its axis, c is on the axis and $\text{dist}_1^2 = \text{dist}_3^2$ is a one-variable quadratic. Else use the dispatcher.	2-active closed
2 parallel + perpendicular	2	Between the lines: $r = w/2$, c not unique. Outside: take S at $x = x_S > h$. The far line and L_3 are active: $r = (x_S + h)/2$, $x_c = (x_S - h)/2$, plane $z = 0$.	closed

Table 1: Cases of the problem's Figure 1 and two more. 2-active closed: the three-active subcase is open.

3 Case Analysis

Symbols in Table 1: "n meet" counts intersection points, as in the problem's Figure 1. δ is the distance of a skew pair, and a parallel pair has gap $w = 2h$ and midline M. We put a parallel pair L_1, L_2 at $x = \mp h$, $z = 0$, along y, and write $L_3 = \{a_3 + t\,e\}$. A *perpendicular* line is normal to the plane of two meeting or parallel lines and meets it at S. For a skew pair it is parallel to their common perpendicular. For lines crossing at P with angle θ, b is the bisector toward S and $v = S - P$. Also $q(z) = \min_y \text{dist}_3^2 - z^2$, with z the height of c above the parallel pair's plane.

4 Verification and Open Problems

We compared Table 1 with a second-order cone program (SOCP), $\min r$ s.t. $\lvert P_i(c - a_i)\rvert \le r$. The inputs were 20 presets and 1,000 random instances per case (18 cases, seed 20261107). An instance passes if three values are at most 10^{-6}: the relative error $\lvert r_{\text{closed}} - r_{\text{socp}}\rvert / \max(1, r_{\text{socp}})$, the touch residual, and the excess of the center over r_{socp}. The code (commit 6337385) ran on the Cradle cluster. A first run exposed an unstable plane normal for coincident lines, fixed in v2. All 18 cases pass, with 0 closed-form failures and 0 solver failures. The largest relative error is 9.0×10^{-9} and the largest touch residual is 2.6×10^{-10}. The solver returned optimal on 18,015 of 18,020 rows, and the other 5 solved on a retry at tolerance 10^{-8}. Verification covers only inputs with a closed form, 14,833 of 18,020 rows. The other 3,187 rows have no closed form and ran on the SOCP only. Closed-form coverage: 296/1,002 (2 meet, 1 cross), 446/1,001 (1 meet, over), 216/1,001 (1 meet, over/under), 289/1,002 (all skew, 713 SOCP only), 573/1,001 (skew perpendicular).

Open problems and future work. The general skew case with three active lines is open. Its eliminant degree and Galois group decide whether radicals suffice, as for the 12 common tangents of four unit balls [2]. We leave that degree to future work. Theorem 1 fails for four lines, e.g. lines through the vertices of an inscribed regular tetrahedron, each perpendicular to its radius. Its center is optimal with $r^* = R$, but the touch points are not coplanar.

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References

[1] B. K. Bhattacharya et al. Optimal algorithms for some intersection radius problems. *Computing*, 52(3), 1994.

[2] I. G. Macdonald, J. Pach, and T. Theobald. Common tangents to four unit balls in $\mathbb{R}^3$. *Discrete Comput. Geom.*, 26(1), 2001.